Learned selection for exact Feynman integral reduction with Kira

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Abstract

We study three opportunities for lightweight machine learning to reduce the cost of integration-by-parts reduction: seed selection, row selection during elimination, and target-dependent equation selection. Exact algebra determines the reduction coefficients, and incomplete selected systems trigger a native fallback. A seed classifier trained on constructed integral relations gives sufficient equation sets for 94.515% of 690,382 test targets. In a separate supplied-equation comparison, seed selection reduces single-target calculation time by 8.6%. Learned row selection gives no reliable improvement in complete native reductions, so we omit it from the retained approach and keep seed and equation selection for their separately validated domains. Selecting equations before coefficient construction reduces preparation time by 36.2% and complete native time by 5.7% on new parameter profiles within four tested graph structures. A frozen filter increases time by 25.2% on new scattering families, including symbolic kinematics, which limits this result's domain. Separately, exact recurrences and reuse give a 543.8-fold speedup for a restricted vacuum family; this is a combined mathematical and implementation gain. The measurements support selective use of learned decisions and retain native choices where their complete cost is lower.

1 Introduction

Higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes contain loop-momentum integrals. Their number and algebraic complexity make reduction to master integrals an important part of precision calculations in quantum field theory [1]. Integration-by-parts (IBP) identities relate integrals with different propagator powers [2]. Kira implements the Laporta approach, organizes these relations into linear systems, selects equations and expresses requested integrals through a master basis [3, 4]. This reduction supplies rational coefficients; evaluating the master integrals remains a separate problem.

Kira already uses optimized seeding and equation selection, so a learned decision must improve upon an existing reduction strategy [5]. Earlier work has used machine-assisted search to discover seed-priority functions [6]. Here we study inexpensive models that guide three decisions during a reduction: which seeds supply identities, which allowed source row eliminates the next integral, and which identities need coefficient construction for the requested targets. We call these System 1 models to specify their role as fast decision rules; the term imposes no particular neural architecture.

Our criterion is complete calculation time, including model evaluation and failed attempts. We first define the reduction and the three decision points, then describe each model and its measured results. We retain the seed and equation models in the domains supported by their separate comparisons and omit learned row selection. The combined-results section states the limits of this choice and distinguishes learned gains from exact mathematical reuse. Appendices give the derivations, model details and measurement protocol needed to assess the results.

2 Kira reduction and the three decision points

We first define the notation used in this paper. An integral family fixes inverse propagators and the external scalar products, while its index vector specifies their powers [1, 4]:

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{i=1}^{N} D_i^{-a_i}, \qquad D_i = q_i^2 - m_i^2 + i0. \tag{1}$$

Here N is the number of inverse propagators, including auxiliary entries. The quantity L is the number of loops, ℓ_h is a loop momentum, $d = 4 - 2\epsilon$ is the regulated dimension, and ϵ is the dimensional regulator. The vector q_i is a linear combination of loop and external momenta, m_i is a mass, and $i0$ specifies the Feynman prescription. The vector $\boldsymbol{x}$ contains independent momentum invariants and squared masses. The integer a_i is the power of denominator i: a positive value places it in the denominator, zero removes it, and a negative value gives a numerator factor. Auxiliary inverse propagators complete the independent scalar products when necessary. A sector specifies which indices are positive [4].

A seed is an index vector $\boldsymbol{\sigma}$ at which one applies a total derivative. An IBP operator differentiates with respect to a loop momentum and contracts with a loop or external momentum. Under dimensional regularization, the integrated total derivative gives a relation [2]

$$0 = \int \prod_{h} \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_i D_i^{-\sigma_i} \right) = \sum_j A_{\boldsymbol{\sigma},o;j}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(j)}}. \tag{2}$$

The label o identifies the derivative and vector v, j labels the integrals produced, and $A_{\boldsymbol{\sigma},o;j}$ is their coefficient. Appendix A expands this identity and explains how a nonzero pivot eliminates an integral. A seed generally supplies several equations, so selecting seeds and selecting individual equations are different decisions.

Kira generates relations in a bounded index region, applies symmetry and equation-selection procedures, and performs forward elimination and back substitution [4, 5]. Finite-field calculations can identify useful equations or support rational reconstruction, while symbolic reduction manipulates rational functions directly [4, 7]. For a target expression $T = \sum_u c_u I_{\boldsymbol{a}^{(u)}}$, the desired result is

$$T = \sum_{m=1}^{n_M} r_{Tm}(\boldsymbol{x}, d) M_m, \qquad r_{Tm} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{3}$$

Here c_u specifies the requested expression, n_M is the number of prescribed masters, and M_m denotes master m. We check completion against that prescribed basis. Agreement with a separately obtained exact reduction tests the coefficients; it does not establish that every chosen seed is necessary.

The first intervention reduces the seed set before equation generation. The second selects a source row among permitted elimination actions while preserving Kira's integral order and protected masters. The third selects seed–operator pairs before constructing their symbolic coefficients. Each choice can save different work and introduces its own prediction cost. We therefore test complete reductions as well as the affected stage.

3 Integral data and comparison design

We constructed targets from checked master relations and exact source identities, retaining the seeds, operators, coefficients and reverse construction steps. If the identity rows are $E_q = \sum_i A_{qi} I_i = 0$, the construction has the form

$$T = \sum_m r_m M_m + \sum_q \lambda_q E_q, \qquad \lambda_q, r_m \in \mathbb{Q}(\boldsymbol{x}, d). \tag{4}$$

Expanding the right-hand side produces the stored target expression and its known master expansion. Appendix B gives the algebraic check. Such a construction can produce a sum of integrals; a single indexed target requires the expanded nonmaster terms to have that form. The construction records a sufficient path without proving minimum cost.

The resulting collection contains 17,100 family instances, 57,979 family-local master entries and 2,301,272 targets. The grouped division contains 1,610,890 training targets in 11,963 families and 690,382 test targets in 5,137 families. Related mass profiles stay in one split across loop orders. The collection spans $L = 1, \ldots, 12$, but many instances are parameter variations of a small set of graph structures. In particular, the added higher-loop data use the restricted vacuum family defined in Appendix D. The test set contains 148,494 single integrals and 541,888 integral expressions. Appendix F defines the integral families and their kinematics, and Appendix G explains the master-basis checks. These counts describe our constructed dataset; they do not count distinct scattering topologies.

The original seed and row models each received twenty additional training passes over the expanded training data, retaining the earlier model and optimizer states. Test families remained outside parameter updates. Earlier stages had evaluated some of the older reserved profiles, so the expanded test is a grouped parameter-transfer test rather than a wholly new topology test. Later native comparisons use separate development, validation and reserved physical profiles, with policies fixed before the final timing comparisons.

Every runtime comparison uses identical requested targets and a common master basis for its competing methods. Complete time includes preparation, inference, exact algebra, result checks and fallback as specified in Appendix E. Most native tests use the same modified Kira 3.1 executable for learned and native choices. The cumulative comparison separately uses the saved original executable. We report five-repeat median ratios and descriptive intervals over physical groups; these quantify the recorded cases and do not establish performance over all possible integral families.

4 Model 1: selecting seeds

The seed model receives the target terms, the family definition, the prescribed master indices and a finite candidate seed list. It uses shared 32-component term representations, all ordered pairs of distinct terms, family and basis summaries, and a candidate-dependent attention average. Shared aggregation and attention follow the constructions of Refs. [8, 9]; Appendix C specifies our implementation. The model has 28,097 adjustable parameters and assigns each seed its own sigmoid score,

$$p_b = \frac{1}{1 + e^{-g_\theta(T,F,\mathcal{M},\boldsymbol{\sigma}_b)}}, \qquad \widehat{\mathcal{S}}(T) = \{\boldsymbol{\sigma}_b \in \mathcal{C} : p_b \geq \tfrac{1}{2}\}. \tag{5}$$

Here F is the family, $\mathcal{M}$ is the master list, $\mathcal{C}$ is the candidate pool and θ denotes fitted parameters. The score describes membership in a recorded sufficient seed set. Scores for different seeds need not sum to one, and a target can require several selected seeds. Training uses target-averaged binary cross-entropy with AdamW [10]; Appendix C derives the loss from the label likelihood. No target master coefficients or future reduction steps enter prediction.

On all 690,382 test targets, precision is 95.750%, recall is 96.892%, and F_1 is 96.318%. Complete predicted sets match the recorded sets for 84.332% of targets. The equations supplied by the predicted seeds suffice for 652,517 targets, or 94.515%, in the Kira sufficiency test. The full pool suffices for all targets in that test. These results measure different questions: agreement with one stored set and the ability to complete a reduction. Appendix E defines the metrics. Candidate accuracy alone obscures the frequent zero labels, a known issue for imbalanced classification [11].

The separate timing experiment reduces constructed expressions from existing identity pools in twelve held-out one-loop box instances. It uses unchanged weights with threshold 0.01, fixed

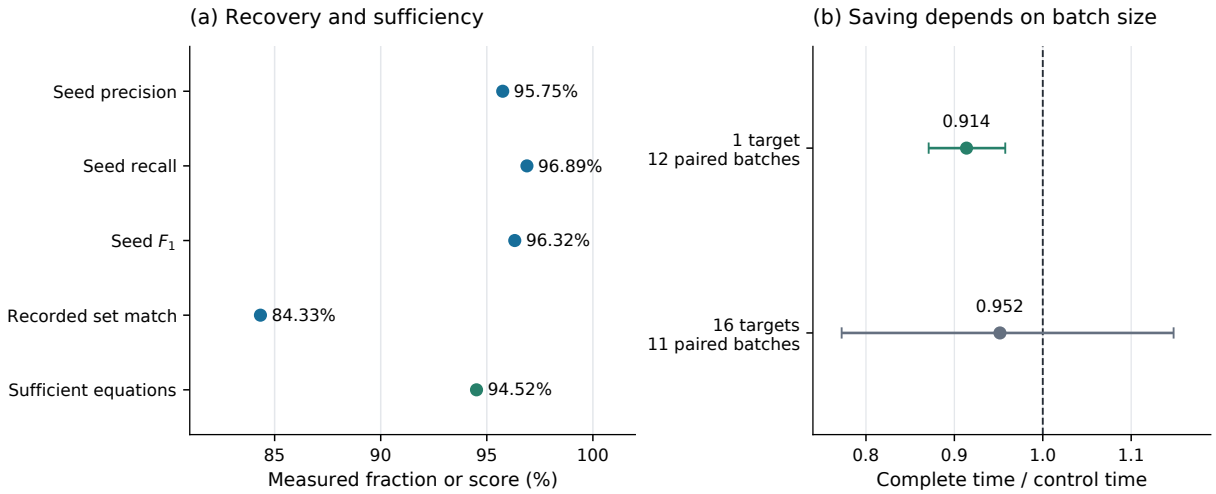


Figure 1: Seed selection separates label recovery, algebraic sufficiency and elapsed time. The left panel reports the frozen expanded-data test. The right panel shows the separate supplied-equation comparisons with the full-pool control. Points and descriptive 95% intervals show that the single-target saving does not extend reliably to the larger batch. The time-ratio axis is explicitly restricted to the range needed to display the intervals; the vertical line marks equal time.

before this reserved comparison; the preceding label and sufficiency tests retain threshold 1/2. Both the learned selector and a distance-based selector receive the same early basis check and full-pool fallback. For one target at a time, the learned/full-pool complete-time ratio is 0.914 with descriptive 95% interval $[0.871, 0.958]$; the ratio to the matched distance control is 0.913 with interval $[0.855, 0.972]$. Thus the measured reduction is 8.6% against the full pool and 8.7% against the distance control. All five repetitions complete for every method in these twelve single-target cases.

For sixteen-target batches, the learned/full-pool ratio is 0.952 with interval $[0.772, 1.148]$, which does not establish a consistent saving. Across the full planned comparison, 345 trials complete exactly, three measured trials are incomplete and twelve repetitions are skipped. The timing summary uses complete paired batches and preserves these failures. This seed result excludes the original construction of the supplied identity pool. In a separate native comparison that includes IBP generation, the later compiled seed route gives a ratio of 1.096 with interval $[0.963, 1.275]$. We retain learned seeds for the supported supplied-equation domain and make no general native speedup claim.

5 Model 2: choosing elimination rows

The row model receives the current coefficient matrix and a finite list of permitted source-row actions. Its 132 features describe the pivot integral, row support, family, matrix size and sampled cancellation counts. A shared neural map produces 32-component action vectors, combines their mean and componentwise maximum, and uses a 64-component hidden layer to assign scores. The network has 10,529 parameters. A softmax converts its scores into a probability distribution over the supplied actions; the highest score selects the next row operation. Appendix C defines the distribution and training objective. Exact arithmetic applies the selected operation, as derived in Appendix A.

Training traces contain one changing target row with fixed source identities. The reserved traces contain 2,558,472 states with more than one action. The model chooses the recorded action in 95.845% of those states, while the local deterministic rule reaches 99.473%. Complete paths agree with the recorded master row at three finite-field samples for 690,302 of 690,382 targets. These

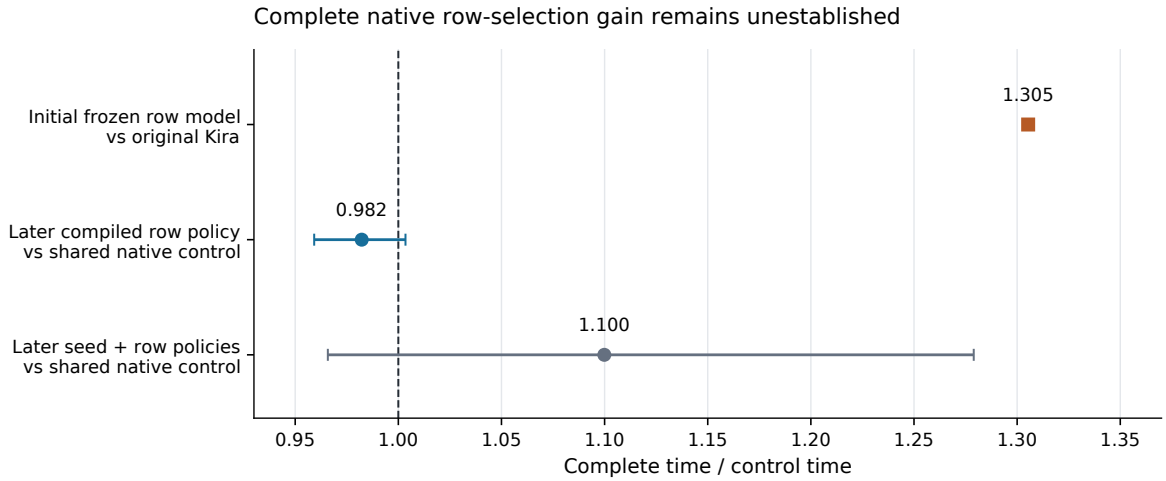


Figure 2: Learned row selection does not establish a reliable complete native speedup. The initial frozen model gives a ratio of 1.305 to stock Kira; its marker has no interval because this summary supplies none. The later comparison uses a shared modified executable and shows descriptive 95% intervals. The row-only interval crosses equal time, and the combined seed–row route gives no established improvement. These are separate comparisons, not successive factors to multiply.

sample checks do not constitute a proof of symbolic equality. The original model takes 17.657 times the local rule's time, including prediction, in this target-row test.

We then inserted the frozen model into Kira's evolving symbolic matrix, preserving the allowed integral order. In 84 exact-checked trials, the complete-time ratio to stock Kira is 1.305. The original training states therefore did not yield a useful native row policy. Subsequent smaller policies use measured immediate and continuation costs, compiled inference and shared finite-field features. The latest row-only complete-time ratio is 0.982 with descriptive interval $[0.959, 1.003]$. This interval includes equal time. Adding the row and seed policies together gives 1.100 with interval $[0.966, 1.279]$ against native choices in the same later comparison.

Figure 2 distinguishes the initial slowdown from the uncertain later result. We omit the second model from the retained reduction method because these complete-time measurements do not establish a reliable gain. Kira continues to choose elimination rows. This decision preserves the exact row-model tests as evidence and avoids treating agreement with a recorded path as evidence of faster algebra.

6 Model 3: selecting equations before coefficient construction

The equation model scores a seed–operator pair for the requested target list before Kira constructs that identity. Its 17 features describe loop and external-momentum counts, target count, seed sector, additional propagator powers, numerator rank, distances from target limits and the operator index. These features require no symbolic equation coefficients. A classification tree estimates membership in the useful support of a recorded native reduction, following the tree framework of Ref. [12]. Accepted pairs enter Kira's native coefficient construction. Symmetry relations and the prescribed basis retain their native treatment.

We trained trees on 153,124 candidate equations, with 12,270 positive labels, from fourteen batches in eight physical profiles. The structures include a two-loop sunrise, a planar double box, a three-loop vacuum and a three-loop ladder. Candidate trees have maximum depths eight and twelve, with 295 and 1,335 nodes. Validation selects depth twelve for the double box and depth eight for vacuum and ladder cases, at a retention threshold of 0.01. The recorded 21-target sunrise domain retains the earlier equation policy. Validation requires lower complete and preparation

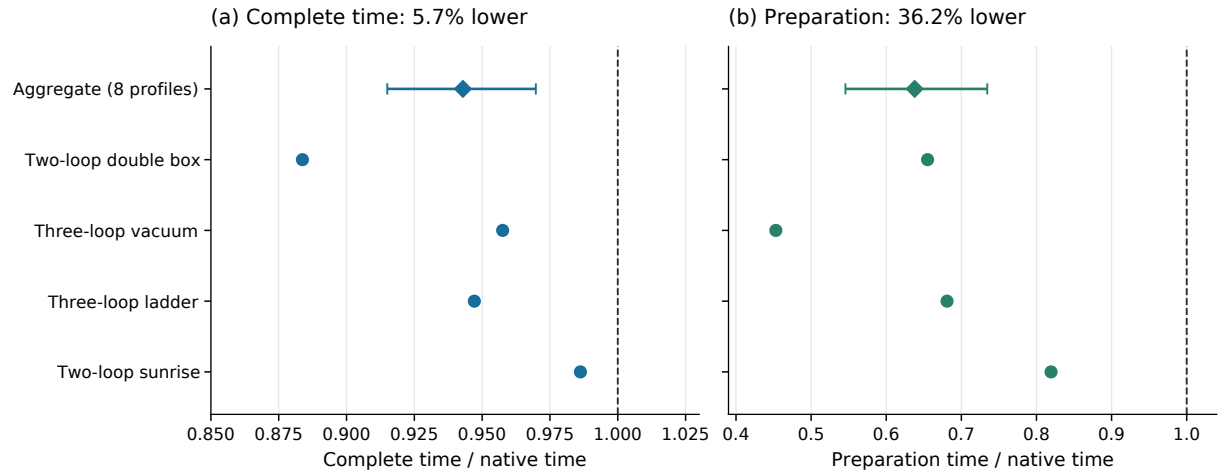


Figure 3: Earlier equation decisions save preparation work, while other stages limit the complete-time gain. Aggregate points show profile-weighted ratios and descriptive 95% intervals over eight profiles. Each graph-specific point summarizes two profiles and carries no separately claimed interval. The two panels use different, labelled axis ranges to resolve complete-time changes and larger preparation changes. Both show the equal-time reference.

times without fallback before activating an early filter. Additional seed and row policies fail the corresponding composition criteria, so these native equation tests retain native seed and row choices.

The independent comparison uses eight new parameter profiles and fourteen target batches. All masses and momentum invariants take specified rational values while d remains symbolic, so coefficient comparisons occur in $\mathbb{Q}(d)$. Five alternating repetitions of four methods give 280 complete workflows. The native, early-filter and distance-control arms use the same executable and shared runtime improvements; the fourth arm uses the preceding release. This design measures model-dependent choices without assigning shared executable improvements to the model.

The deployed equation policy gives a complete-time ratio of 0.943 with descriptive 95% interval $[0.915, 0.970]$ against native choices. Its equation-preparation ratio is 0.638 with interval $[0.546, 0.734]$. These correspond to 5.7% less complete time and 36.2% less preparation time. Preparation includes modular generation, equation selection and solving, and symbolic equation regeneration. Figure 3 shows why a larger stage saving produces a smaller complete saving. All 280 workflows and eight applications with only a prescribed basis pass the independent exact audit, covering 94,970 coefficient comparisons. Every method completes on its first attempt.

We next froze the depth-twelve tree and tested previously unseen scattering structures: a massless nonplanar two-loop box, a massless one-loop pentagon, and a massless box with one independent external virtuality. Appendix F gives the denominator momenta and invariant conventions; the nonplanar family follows the massless limit of the corresponding family in Ref. [4]. Four rational-kinematic batches keep d symbolic; three additional batches keep their independent invariants symbolic. The latter reductions occur in $\mathbb{Q}(d, s, t)$, $\mathbb{Q}(d, s_{12}, s_{23}, s_{34}, s_{45}, s_{51})$, and $\mathbb{Q}(d, s, t, v)$, respectively. Here s, t are four-point invariants, $s_{i,i+1}$ are adjacent five-point invariants, and v is the external squared momentum.

The seven batches contain 88 targets and give seventy comparison workflows. Every filtered workflow requires native fallback. The complete-time ratio is 1.252, or 25.2% more time. All workflows pass exact comparisons, including 5,600 coefficient-pair checks. Enlarged seed regions agree for all seven cases, and fresh rational specializations agree for the three symbolic cases. These checks support the returned reductions while the measured slowdown prevents activation of the filter in these new domains. Figure 4 reports each case rather than extrapolating the earlier gain.

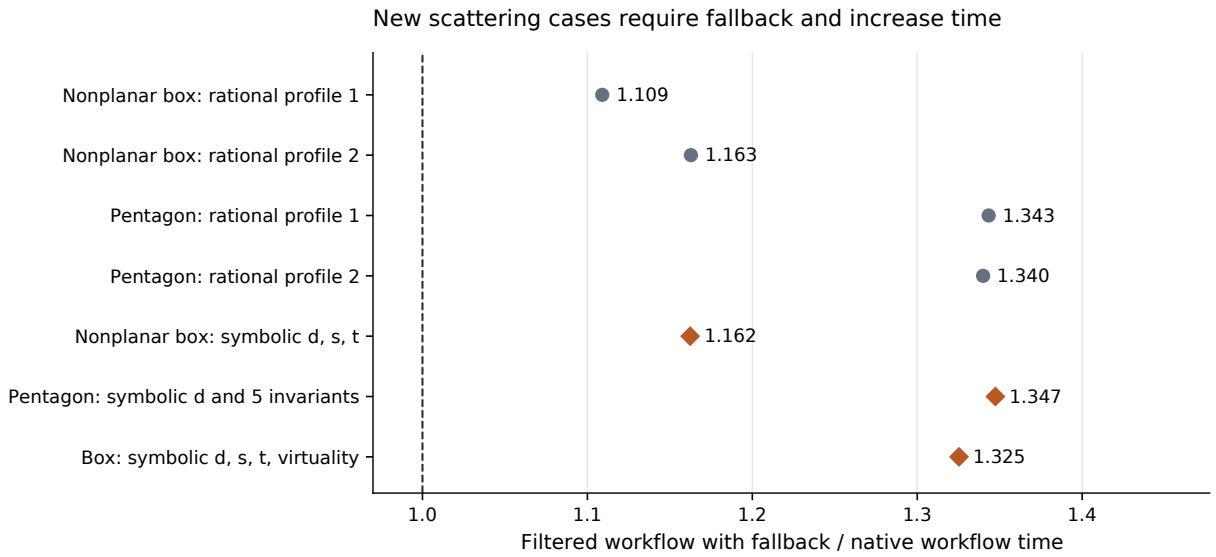


Figure 4: The frozen equation filter loses its timing benefit on new scattering families. Each marker divides the median filtered-workflow time by the native median for one case, including fallback. Five repetitions contribute to each median. No per-case confidence interval is inferred from those five repetitions. All 35 filtered workflows fall back to native equations, and the geometric mean ratio is 1.252.

7 Overall results and retained method

The retained learned methods are seed selection and target-dependent equation selection. We omit learned row selection and keep Kira's native elimination choices. The two retained models operate in their separately validated domains: learned seeds for supported supplied-equation requests, and the equation policy for supported native requests. Their measured gains cannot be multiplied because the comparisons use different inputs and include different work. The native equation experiment does not establish a gain from activating both models together; validation rejected that composition. For an unsupported domain, the application uses native choices or retries them after an incomplete experimental selection.

We separately measured the cumulative effect of exact mathematical reuse and program changes in the restricted vacuum family. Appendix D derives the shared variable $z = L(d-2)$, the decreasing-power recurrences and an independent Taylor check. The final implementation evaluates these recurrences with a fixed validated ordering, reuses intermediate coefficients across targets and loop orders, uses exact arithmetic and checks the resulting coefficients. Kira remains a fallback if the direct check fails. No additional learned ordering supplies an established gain in this final specialized route.

Eight fresh groups contain three mass profiles each, with two, three or four distinct massive denominators, and request loop orders one through twelve. Five repetitions compare the saved original Kira 3.1 executable, a later full-pool control and the final specialized route on identical requests. All 120 workflows pass exact checks. The measured final/original ratio is 0.001839, equivalent to a speedup of 543.815 with descriptive 95% interval $[488.134, 605.064]$. An independent audit checks 27,960 nonzero target–master coefficient polynomials. The final route makes no Kira calls on these successful requests, while each baseline arm makes 1,440 calls across its trials.

The cumulative benchmark generates and supplies Euler and mass partial-fraction identities within the timed workflow. It does not generate a general native high-loop IBP system. Earlier checks at $L = 13, 16, 24, 32$ use the same restricted family, whose recurrence proof permits arbitrary positive integer L. These results therefore concern a stated mathematical structure rather than general high-loop scattering. The broader symbolic tests in Section 6 establish exact

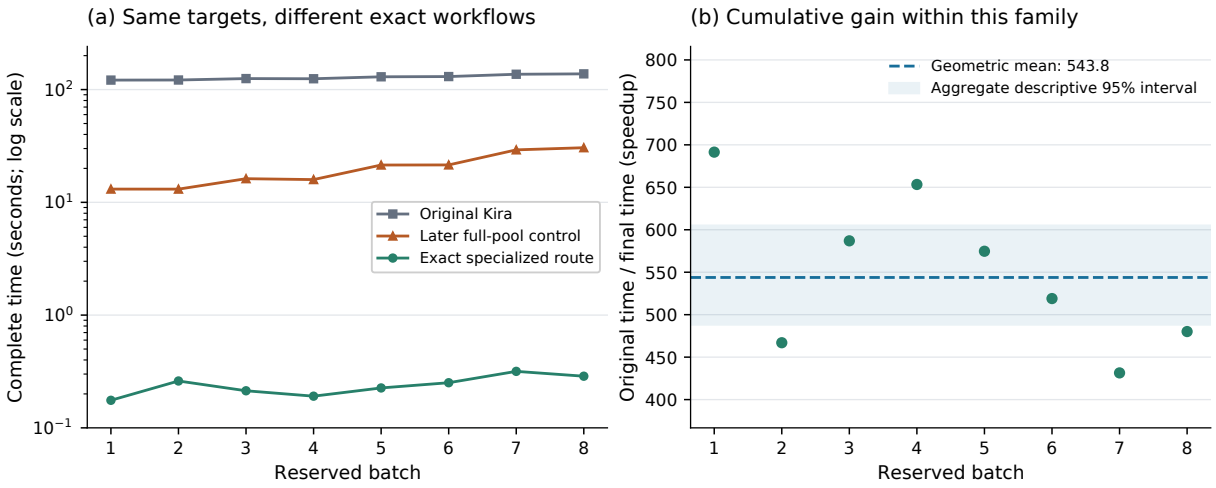


Figure 5: Exact recurrence reuse gives the large cumulative gain in the restricted vacuum family. Left: absolute complete-time medians on a logarithmic axis, including fresh-worker startup, checks, output saving and reading. Right: original-baseline median divided by final median for each batch. Every batch requests three mass profiles and loop orders 1–12. The aggregate 543.8-fold speedup measures the complete specialized change; it does not isolate a learned-model effect.

completion with native fallback, but give no further learned speedup.

8 Conclusion

The measurements show that learned decisions help only when their complete cost is lower than the work they avoid. Seed selection supplies sufficient equations for 94.515% of the expanded test targets and reduces time by 8.6% in the separate single-target supplied-equation comparison. Target-dependent equation selection reduces native preparation time by 36.2% and complete time by 5.7% within the tested graph structures. Learned row selection gives no reliable complete-time improvement, so we omit that second model and retain the other two methods with their stated domain restrictions.

New scattering structures and symbolic invariants expose a transfer limit: the frozen equation filter requires fallback and increases time by 25.2%. The retained application therefore uses native choices in those domains. The separate 543.8-fold cumulative speedup comes from exact recurrences, shared reductions and implementation changes in a restricted vacuum family. General native multiloop speedups and a beneficial simultaneous seed–equation composition remain unestablished by these measurements. Small bubble and tadpole systems also admit minimum-operation proofs within a fixed source pool, as derived in Appendix H. The corresponding minima are seven coefficient operations for each system when initial equation selection is allowed. These finite examples do not prove global optimality for the learned reductions. The exact checks and complete-cost comparisons make those limits part of the result.

A IBP relations and exact elimination

Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k^\mu = \partial/\partial \ell_{k\mu}$. Applying the product rule to the integrand in Eq. (2) gives

$$\partial_k^\mu (v_\mu P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} + v \cdot \partial_k P_{\boldsymbol{\sigma}}, \tag{6}$$

$$v \cdot \partial_k P_{\boldsymbol{\sigma}} = -\sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{7}$$

For $v = \ell_h$, $\partial_k \cdot v = d\delta_{kh}$; for an external momentum it is zero. The integrated derivative vanishes in a convergent domain and defines the dimensionally regulated identity by analytic continuation [2, 1]. This use of dimensional regularization includes its treatment of scaleless sectors.

With a complete scalar-product basis, the scalar numerator $v \cdot \partial_k D_i$ can be written as $u_i + \sum_j v_{ij} D_j$. The coefficients u_i and v_{ij} depend on masses, invariants and the chosen momentum routing. Substituting in Eq. (7) gives

$$0 = \delta_v I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma}+\boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma}+\boldsymbol{e}_i-\boldsymbol{e}_j}, \tag{8}$$

where $\delta_v = \partial_k \cdot v$ and $\boldsymbol{e}_i$ increments index i by one. Thus differentiation produces linear relations with shifted integer indices. Auxiliary denominators or separately supplied algebraic relations handle dependent representations; the restricted family in Appendix D uses explicit mass partial fractions.

Suppose row q of such a system satisfies $\sum_j A_{qj} I_j = 0$ and $A_{qk} \neq 0$. Solving this row for the pivot integral gives

$$I_k = -\sum_{j \neq k} \frac{A_{qj}}{A_{qk}} I_j. \tag{9}$$

For a second identity row, or a target coefficient row r, subtract (A_{rk}/A_{qk}) times the zero identity. The resulting coefficients are

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{10}$$

The target retains its value because the subtracted expression is zero. A different permitted source row changes the intermediate expressions but preserves this argument. Iterating until every unprotected column vanishes yields Eq. (3). A nonzero pivot condition applies over the rational-function field; special kinematic points where denominators vanish require separate treatment.

B Reverse construction and the meaning of reference labels

For each checked source identity E_q, its coefficients sum to zero when evaluated as integrals. Substituting these identities into Eq. (4) gives

$$T - \sum_m r_m M_m = \sum_q \lambda_q E_q = 0. \tag{11}$$

This establishes the master coefficients of the constructed expression. Expanding each E_q in indexed integrals and collecting equal indices defines the input target. The reverse operations, source rows and their seed–operator labels supply a reproducible path back to the masters. The coefficients λ_q and r_m can be rational functions; constants form a special case. Mass partial-fraction identities have no IBP seed label.

Adding another zero identity changes the construction history without changing the integral expression. This explains why a recorded sufficient seed set need not be unique. It also explains why a reference elimination path need not minimize the number of operations. Our binary labels mean membership in one checked reference set. Our action labels mean the next recorded step. Neither label supplies an unrestricted optimality certificate. The target's final master coefficients and later construction steps remain outside the features used at inference.

C Model specification and training objectives

For seed selection, a fixed signed token-count map encodes coefficient expressions into 64 components, including individual tokens and consecutive groups of two and three tokens. Different groups can share a component, so this representation does not establish symbolic equality. Integral indices, complexity counts and the stored term position supplement these features. A shared hyperbolic-tangent map gives a term vector $\boldsymbol{h}_u \in \mathbb{R}^{32}$. Family symbols, master indices and candidate seed indices give 32-component family, basis and seed representations.

The ordered-pair summary is

$$\boldsymbol{h}_{\mathrm{pair}} = \frac{1}{n_T(n_T-1)} \sum_{u \neq v} \psi_\theta([\boldsymbol{h}_u, \boldsymbol{h}_v, \boldsymbol{h}_u - \boldsymbol{h}_v, \boldsymbol{h}_u \odot \boldsymbol{h}_v]), \tag{12}$$

where n_T is the number of terms, square brackets concatenate vectors, $\odot$ multiplies components and ψ_θ returns 32 components. For $n_T = 1$, the summary is zero. This definition covers ordered pairs, not all larger term subsets. The candidate vector $\boldsymbol{u}_b$ weights target terms through scaled dot-product attention [9]:

$$w_{bu} = \frac{\exp(\boldsymbol{u}_b \cdot \boldsymbol{h}_u/\sqrt{32})}{\sum_v \exp(\boldsymbol{u}_b \cdot \boldsymbol{h}_v/\sqrt{32})}, \qquad \boldsymbol{v}_b = \sum_u w_{bu} \boldsymbol{h}_u. \tag{13}$$

The output map joins the mean and maximum term vectors, pair vector, family vector, basis vector, seed vector and weighted-term vector. A 64-component hidden layer produces the logit g_θ in Eq. (5). Shared maps and aggregation follow the set-based construction of Ref. [8]; stored positions retain term order in this implementation. Candidate reordering only reorders the outputs.

If $y_b \in \{0, 1\}$ is recorded seed membership, a Bernoulli model assigns label likelihood $p_b^{y_b}(1 - p_b)^{1-y_b}$. Taking its negative logarithm gives $-y_b \log p_b - (1 - y_b) \log(1 - p_b)$. Averaging within each candidate list and then across targets gives the implemented loss

$$\mathcal{L}_s = -\frac{1}{|\mathcal{D}_s|} \sum_{T \in \mathcal{D}_s} \frac{1}{|\mathcal{C}(T)|} \sum_{b \in \mathcal{C}(T)} [y_b \log p_b + (1 - y_b) \log(1 - p_b)]. \tag{14}$$

Here $\mathcal{D}_s$ is the training-target set. This objective treats each target equally and does not claim that true seed dependencies are statistically independent. Training uses AdamW [10] with learning rate 10^{-3}, weight decay 10^{-4}, gradient norm limit one and up to 32 targets per update. The threshold 1/2 stays fixed during the expanded label test.

For row selection, let $\mathcal{A}$ be the current allowed action list and $\boldsymbol{f}_a \in \mathbb{R}^{132}$ its feature vector for action a. The encoder is $\boldsymbol{h}_a = \tanh(W\boldsymbol{f}_a + \boldsymbol{b})$ with 32 components. Concatenating $\boldsymbol{h}_a$ with the mean and componentwise maximum over $\mathcal{A}$ gives the input to a 64-component hidden layer. Its scalar output g_a defines

$$p(a \mid A, \mathcal{A}) = \frac{e^{g_a}}{\sum_{b \in \mathcal{A}} e^{g_b}}, \qquad \mathcal{L}_r = -\frac{1}{|\mathcal{D}_r|} \sum_{(A, \mathcal{A}, a^*) \in \mathcal{D}_r} \log p(a^* \mid A, \mathcal{A}). \tag{15}$$

Here A is the matrix, a^* is the recorded action and $\mathcal{D}_r$ contains states with an actual choice. The denominator normalizes the probabilities; taking the negative logarithm of the categorical reference likelihood gives the loss. Single-action states have zero loss. Training uses the same AdamW learning rate, decay and norm limit as seed training, with 256 states per update. Finite-field trial updates supply cancellation features; a modular residue's numerical size does not represent a real coefficient magnitude.

The later native row policies change the fitted target to the cost caused by a candidate source. An immediate-cost fit predicts the logarithm of measured exact insertion time. A continuation

fit instead counts the remaining forward insertions after the candidate choice, followed by native choices. If insertion j uses n_j source entries, its specified estimate is $1 + 2n_j$: one division and one multiplication and subtraction per entry. The continuation label is therefore $C(S, q) = \sum_j (1 + 2n_j)$. The later tree fits

$$y_q = \frac{C(S,q) - C(S,0)}{\max\{1, \max_b C(S,b) - \min_b C(S,b)\}}, \qquad \widehat{y}_q = h(\boldsymbol{x}_q - \boldsymbol{x}_0), \tag{16}$$

where 0 denotes Kira's native source and $\boldsymbol{x}_q$ contains eighteen current-row and overlap features. Weighted squared error gives each training family equal weight. Validation selects the tree and the required predicted improvement, and exact coefficients must agree before a measured continuation enters training. This count describes coefficient operations, while complete timings include the algebraic work inside each coefficient operation. Neither training objective proves a globally minimum path.

The equation tree uses the seed, operator and target features described in Section 6. Its seventeen inputs are the loop count, independent external-momentum count, index count, logarithm of target count, number of positive seed indices, additional seed powers, seed numerator rank, excess additional powers and rank over the target maxima, whether the target list requests the seed sector, the corresponding two excesses within that sector, minimum seed–target index distance, whether the seed itself is a target, operator index, and the two overall target maxima. The index distance is $\sum_i |\sigma_i - a_i|$. Additional powers and numerator rank are $\sum_i \max(\sigma_i - 1, 0)$ and $\sum_i \max(-\sigma_i, 0)$, respectively. Coefficient-expression features are excluded, so prediction can precede coefficient construction.

Training gives each physical profile equal total weight by assigning each row weight $1/(n_b B_g)$, where n_b is its batch size and B_g is the number of training batches in its profile. At a node, let w_+, w_- be the positive and negative total weights. Half the weighted binary Gini impurity is

$$H = \frac{w_+ w_-}{w_+ + w_-}. \tag{17}$$

Indeed, with $p = w_+/(w_+ + w_-)$, the weighted Gini expression is $(w_+ + w_-)[1 - p^2 - (1-p)^2] = 2H$. Each proposed split reduces H by the sum of its two children's values. The fit tests 25 feature quantiles between 0.02 and 0.98, rejects children below the stated population limit, and selects the largest positive reduction. This is the classification-tree construction of Ref. [12] with our bounded set of candidate splits.

A leaf with n_+ positive rows among n rows returns

$$p_{\text{leaf}} = \frac{n_+ + 1/2}{n + 1}. \tag{18}$$

This definition adds one half-count to each class and keeps a finite score even for a pure leaf. The acceptance condition is $p_{\text{leaf}} \geq 0.01$. These scores estimate recorded support membership; they do not measure a calibrated probability of reduction success. The depth-eight and depth-twelve candidates require at least 24 and eight rows per child, respectively. Classification-tree splits and pruning principles follow Ref. [12]; the specified depths, features and thresholds define our fitted policies. Support-membership scores do not certify that a selected system is complete. A prescribed-basis check and native fallback therefore remain part of each application.

D Exact reuse for the restricted vacuum family

To state the domain of the cumulative comparison, define a Euclidean vacuum family with positive, distinct squared masses $\mu_j = \rho_j \mu$:

$$V_{\boldsymbol{b}}^{(L)} = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{\pi^{d/2}} \frac{1}{\ell_1^2 \dots \ell_L^2} \prod_{j=1}^{R} (Q^2 + \mu_j)^{-b_j}, \qquad Q = \sum_{h=1}^{L} \ell_h. \tag{19}$$

Here b_j is a nonnegative integer, $B = \sum_j b_j \geq 1$, R is the number of massive denominators, μ is a common squared-mass scale and ρ_j is a positive mass ratio. Each massless denominator has power one. All massive denominators share Q, so this family has special algebraic dependencies. Its prescribed masters are $M_j^{(L)} = V_{\boldsymbol{e}_j}^{(L)}$. The comparison uses $R = 2, 3, 4$ and $\mu = 1$ in the selected units.

Let f be the integrand without the measure. The simultaneous Euler derivative gives $\sum_h \ell_h \cdot \partial_{\ell_h}(\ell_i^2)^{-1} = -2(\ell_i^2)^{-1}$ for each massless factor. Also, $\sum_h \ell_h \cdot \partial_{\ell_h} Q^2 = 2Q^2$. Therefore

$$\sum_h \partial_{\ell_h} \cdot (\ell_h f) = \left[Ld - 2L - 2\sum_j b_j \frac{Q^2}{Q^2 + \mu_j} \right] f. \tag{20}$$

Use $Q^2 = (Q^2 + \mu_j) - \mu_j$ and integrate the derivative as in Appendix A. This gives

$$0 = [L(d-2) - 2B] V_{\boldsymbol{b}}^{(L)} + 2\sum_j b_j \mu_j V_{\boldsymbol{b}+\boldsymbol{e}_j}^{(L)}. \tag{21}$$

For two present massive denominators, the integrand identity $(Q^2 + \mu_i) - (Q^2 + \mu_j) = \mu_i - \mu_j$ separately gives

$$(\mu_i - \mu_j) V_{\boldsymbol{b}}^{(L)} = V_{\boldsymbol{b}-\boldsymbol{e}_i}^{(L)} - V_{\boldsymbol{b}-\boldsymbol{e}_j}^{(L)}. \tag{22}$$

This second relation is an algebraic partial fraction, so it introduces no additional IBP seed.

Define $z = L(d-2)$ and $W_{\boldsymbol{b}}^{(L)} = \mu^{B-1} V_{\boldsymbol{b}}^{(L)}$. The factor of μ gives W the masters' mass dimension. Substituting in Eqs. (21) and (22) removes the common scale. If $\boldsymbol{C}_{\boldsymbol{b}}(z)$ is the vector of master coefficients of $W_{\boldsymbol{b}}^{(L)}$, then

$$\boldsymbol{C}_{\boldsymbol{b}}(z) = \frac{\boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_i}(z) - \boldsymbol{C}_{\boldsymbol{b}-\boldsymbol{e}_j}(z)}{\rho_i - \rho_j}, \qquad b_i, b_j > 0, \tag{23}$$

$$\boldsymbol{C}_{k\boldsymbol{e}_j}(z) = \frac{2(k-1) - z}{2(k-1)\rho_j} \boldsymbol{C}_{(k-1)\boldsymbol{e}_j}(z), \qquad k \geq 2. \tag{24}$$

The initial vector is $\boldsymbol{C}_{\boldsymbol{e}_j} = \boldsymbol{e}_j$. Every recurrence lowers B, so it terminates at the prescribed masters. Its coefficients depend on L only through z. Induction on B shows that they are polynomials in z with rational coefficients at fixed rational ρ_j. One calculation can therefore supply all requested loop orders by substituting $z = L(d-2)$.

We independently check the mixed-denominator reduction through Taylor partial fractions. For a dimensionless auxiliary variable x, write

$$\prod_i (x + \rho_i)^{-b_i} = \sum_{j:b_j>0} \sum_{k=1}^{b_j} \frac{a_{jk}}{(x + \rho_j)^k}. \tag{25}$$

At the pole $x = -\rho_j$, set $t = x + \rho_j$. Multiplying by t^{b_j} and expanding the regular factors shows that the principal-part coefficient is

$$a_{jk} = [t^{b_j - k}] \prod_{i \neq j} (\rho_i - \rho_j + t)^{-b_i}. \tag{26}$$

Here $[t^n]$ extracts Taylor coefficient n and factors with $b_i = 0$ are one. Applying Eq. (24) repeatedly to each pole term gives

$$C_{\boldsymbol{b}j}(z) = \sum_{k=1}^{b_j} a_{jk} \prod_{h=1}^{k-1} \frac{2h - z}{2h\rho_j}, \qquad V_{\boldsymbol{b}}^{(L)} = \mu^{1-B} \sum_j C_{\boldsymbol{b}j}(L(d-2)) M_j^{(L)}. \tag{27}$$

An empty product is one. The Taylor calculation avoids the selected pair-recurrence path and checks every returned polynomial coefficient before substitution. The derivation assumes distinct ρ_j, unit massless powers and the shared momentum Q; it supplies no formula for arbitrary multiloop scattering families.

E Measurement definitions and comparison conditions

For seed labels, let TP count selected reference seeds, FP count selected seeds absent from the reference set, and FN count omitted reference seeds. Precision is $P = \mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$ and recall is $R = \mathrm{TP}/(\mathrm{TP} + \mathrm{FN})$. Defining F_1 as their harmonic mean gives

$$F_1 = \frac{2}{P^{-1} + R^{-1}} = \frac{2PR}{P + R} = \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}. \tag{28}$$

Set agreement instead counts targets whose entire predicted set equals the recorded set. Algebraic sufficiency counts targets that reduce to the prescribed basis with the selected source equations. These definitions explain why the three quantities in Figure 1 can differ.

For timing, let T_{abr} be the measured time for method a, batch b and repetition r. We use $\widetilde{T}_{ab} = \mathrm{median}_{r=1,\dots,5}\, T_{abr}$. For the later native comparisons, if physical group g contains batches B_g, define the equal-group mean logarithmic ratio

$$\overline{\ell} = \frac{1}{G}\sum_{g=1}^{G} \frac{1}{|B_g|} \sum_{b \in B_g} \log \frac{\widetilde{T}_{ab}}{\widetilde{T}_{cb}}, \qquad R_{a/c} = e^{\overline{\ell}}. \tag{29}$$

Here c identifies the control and G is the number of physical groups. Exponentiating the mean log ratio gives the geometric mean; ratios below one indicate lower time. Reversing numerator and denominator changes $\overline{\ell}$ to $-\overline{\ell}$, so speedup is $S = 1/R_{a/c}$ and the time reduction is $100(1 - R_{a/c})\%$. The cumulative study uses one batch per group. The supplied-equation seed comparison instead gives each complete paired batch equal weight; its interval resamples six related kinematic groups. Following bootstrap resampling [13], reported descriptive 95% intervals resample complete groups 10,000 times while retaining paired methods and repetitions. They describe case-to-case variation in these experiments. We do not infer new uncertainty intervals for figures whose source summaries provide only point estimates.

The supplied-equation seed test includes target preparation, inference, early basis checking, reduction, result checking and fallback. It excludes original IBP-pool generation. It rotates methods within each family and target batch on the same node, and includes only complete paired batches in its ratio summary while preserving incomplete and skipped trials in the accounting.

The native equation and transfer tests include input preparation, model loading and evaluation, Kira startup, native IBP generation, elimination, basis and coefficient checks, cleanup and every fallback. Independent reference discovery and the later independent check lie outside the timer. In the equation test, the native and learned arms receive the same executable-level improvements. The eight reserved profiles test new masses and invariants within known structures; the seven transfer cases additionally test new structures and symbolic invariants.

The cumulative test includes fresh-worker startup, input loading, supplied Euler and mass-identity construction, algebra, verification, output saving and complete output reading. Eight groups use maximum massive powers 8, 10, 12 or 14, with two groups at each bound. They request 16, 32, 48 or 64 targets per profile, capped by availability. Each group contains three mass profiles and all loop orders 1–12. Paired methods use the same resources, and their order changes between repetition blocks. Each method recomputes the reduction. Waiting for resources, offline training and the later independent check remain outside all reported production times.

The initial live-matrix comparison uses three repetitions of four methods on seven physical cases, giving 84 workflows. Its complete ratio is the geometric mean of the seven paired median ratios and carries no reported uncertainty interval. The later native implementation comparison uses five repetitions and sixteen batches; it repeats earlier physical profiles to measure the implementation change.

Every quantitative statement in Sections 3–7 reports this study's recorded measurements. The evidence separates the expanded label test, supplied-equation seed comparison, live-row comparison, later native comparison, early-equation comparison, new-family transfer test and direct

cumulative test. This separation prevents a change of baseline or workload from entering a claimed combined model speedup. The archived notes retain the earlier experiments and their negative results.

F Integral families and kinematic conventions

The general integral in Eq. (1) becomes a specific family when its ordered denominator momenta, masses and external scalar products are given. The definitions below cover the families used to construct the training data and the families used in the reported comparisons. A target is an index vector in one of these families or a stated rational linear combination of such vectors. Thus the definitions cover the generated integrals without printing each parameter instance. Unless stated otherwise, $D_i = q_i^2 - m_i^2 + i0$ and the measure is that of Eq. (1). The Euclidean family in Eq. (19) retains its separately stated sign convention.

F.1 Initial one-loop and two-loop families

For the one-loop tadpole, $q_1 = \ell$ and $m_1^2 = \mu > 0$. For the bubbles,

$$\boldsymbol{q} = (\ell, \ell - p), \qquad p^2 = s, \qquad (m_1^2, m_2^2) \in \{(0,0), (\mu,0), (\mu,\mu), (\mu_1,\mu_2)\}. \tag{30}$$

The symbols μ, μ_1, μ_2 denote squared masses, with $\mu_1 \neq \mu_2$ in the unequal-mass case. The triangle routing is

$$\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2), \qquad (p_1^2, p_2^2, (p_1+p_2)^2) = (u_1, u_2, u_3). \tag{31}$$

Expanding $(p_1+p_2)^2$ gives $2p_1 \cdot p_2 = u_3 - u_1 - u_2$. Three massless triangle classes use virtualities $(0,0,s)$, $(0,u_2,u_3)$ and (u_1,u_2,u_3). Two equal-mass classes, with all internal squared masses μ, use $(0,0,s)$ and (u_1,u_2,u_3). Each displayed nonzero virtuality is independent at generic kinematics.

The initial massless boxes use four incoming external momenta, $p_4 = -p_1 - p_2 - p_3$, and

$$\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_1 - p_2 - p_3), \quad s = (p_1+p_2)^2, \quad t = (p_2+p_3)^2. \tag{32}$$

Writing $u_i = p_i^2$, the three classes have $(u_1,u_2,u_3,u_4) = (0,0,0,0)$, $(0,0,0,u_4)$ and (u_1,u_2,u_3,u_4). Expanding the definitions of s, t and p_4^2 gives

$$2p_1 \cdot p_2 = s - u_1 - u_2, \quad 2p_2 \cdot p_3 = t - u_2 - u_3, \quad 2p_1 \cdot p_3 = u_4 + u_2 - s - t. \tag{33}$$

These definitions give thirteen one-loop classes. The initial two-loop massless two-point family uses

$$\boldsymbol{q} = (\ell_1, \ell_2, \ell_1 - \ell_2, \ell_1 - p, \ell_2 - p), \qquad p^2 = s, qquad m_i^2 = 0. \tag{34}$$

The equal-mass sunrise uses $\boldsymbol{q} = (\ell_1, \ell_2, p - \ell_1 - \ell_2, p - \ell_1, p - \ell_2)$, with masses $(\mu,\mu,\mu,0,0)$ and $p^2 = s$. Its last two entries are auxiliary: $a_4, a_5 \leq 0$. The equal-mass vacuum sunset has $\boldsymbol{q} = (\ell_1, \ell_2, \ell_1 - \ell_2)$ and all three squared masses μ. These three classes complete the initial sixteen. Their subsectors include the integrals obtained by removing physical denominators.

F.2 Additional one-loop mass and invariant assignments

The 984 added box families use a more general assignment than the three initial massless boxes:

$$D_i = (\ell - Q_i)^2 - \mu_{\alpha_i} + i0, \quad Q_1 = 0, \quad \mu_0 = 0, \quad \alpha_i \in \{0,1,2,3\}. \tag{35}$$

The nonzero squared masses μ_1, μ_2, μ_3 are independent. Repeated labels impose equal masses. The three external shifts Q_2, Q_3, Q_4 have squared differences

$$\Delta_{ij} = (Q_i - Q_j)^2 = y_{\beta_{ij}}, \quad y_0 = 0, \quad \beta_{ij} \in \{0,1,2,3\}. \tag{36}$$

The nonzero y_1, y_2, y_3 are independent invariants. Expanding a squared difference gives the Gram matrix

$$Q_i \cdot Q_j = \frac{\Delta_{1i} + \Delta_{1j} - \Delta_{ij}}{2}, \qquad i, j \in \{2, 3, 4\}. \tag{37}$$

The construction rejects singular sampled Gram matrices and identifies assignments related by propagator permutations or by relabelling nonzero masses and invariants. A fixed order selects 984 of 1,673 admissible inequivalent profiles after excluding the original massless boxes. This extends the initial collection to 1,000 families with 9,678 family-local masters and 240,472 targets before the higher-loop additions. Each profile gives a symbolic family through its zeros and equality conditions. The single-target seed timing comparison uses twelve reserved instances of these boxes. Its targets follow the reverse construction in Appendix B, rather than the native target enumeration below.

F.3 Native multiloop comparisons

The sunrise comparisons allow independent masses on the same three physical propagators. The other native comparison families use

$$\begin{aligned} \boldsymbol{q}_{\mathrm{DB}} &= (\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \\ &\qquad \ell_1 - \ell_2, \ell_1 - p_3, \ell_2 - p_1), \end{aligned} \tag{38}$$

$$\boldsymbol{q}_{\mathrm{V}} = (\ell_1, \ell_2, \ell_3, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3), \tag{39}$$

$$\boldsymbol{q}_{\mathrm{L}} = (\ell_1, \ell_1 - p, \ell_2, \ell_2 - p, \ell_3, \ell_3 - p, \ell_1 - \ell_2, \ell_2 - \ell_3, \ell_1 - \ell_3). \tag{40}$$

The double box has two loops and seven physical denominators; its last two entries are auxiliary. The vacuum has three loops and six physical denominators. The ladder has three loops and eight physical denominators; its last entry is auxiliary. Auxiliary indices remain nonpositive. The massless double box has $p_i^2 = 0$ and uses $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, $2p_2 \cdot p_3 = -s - t$. This local definition of t differs from the initial box convention because the routing differs.

For the final early-equation test, the sunrise squared-mass vectors are $(2, 7, 11)$ and $(3, 7, 13)$, with $p^2 = -101, -103$. The double-box profiles are $(s, t) = (-101, -103), (-107, -109)$. The vacuum squared-mass vectors are $(1, 1, 1, 1, 1, 11)$ and $(1, 1, 1, 13, 13, 13)$. The ladder has squared masses $(1, 0, 0, 0, 0, 0, 0, 0)$ and $p^2 = -101, -103$. All values use a common squared-momentum unit. These choices keep d symbolic and give coefficients in $\mathbb{Q}(d)$.

The target order starts with the full-sector corner, then adds each single extra propagator power and each available single auxiliary numerator. The construction repeats after removing one, two and three physical denominators, retains at least two, and removes duplicates. The comparison takes 21 sunrise targets and batches of 32 and 128 targets for each other profile. This gives the fourteen batches in Section 6.

F.4 New scattering structures and symbolic invariants

The massless nonplanar box uses the routing of Ref. [4]:

$$\begin{aligned} \boldsymbol{q}_{\mathrm{NP}} &= (\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &\qquad p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \end{aligned} \tag{41}$$

Its first seven entries are physical. Its last two are auxiliary. Here $p_1^2 = p_2^2 = p_3^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$ and $t = (p_1 - p_3)^2$. Expanding these squares gives $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = -t$ and $2p_2 \cdot p_3 = s + t$.

The massless pentagon uses $\sum_{i=1}^{5} p_i = 0$, $p_i^2 = 0$ and

$$\boldsymbol{q}_{\mathrm{P}} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_1 - p_2 - p_3, \ell - p_1 - p_2 - p_3 - p_4). \tag{42}$$

All five denominators are physical. The independent invariants are $(s_{12}, s_{23}, s_{34}, s_{45}, s_{51})$, where $s_{ij} = (p_i + p_j)^2$. Squaring $p_1 + p_2 + p_3 = -(p_4 + p_5)$ gives $2p_1 \cdot p_3 = s_{45} - s_{12} - s_{23}$; the other nonadjacent products follow by cyclic relabelling. The box with an external virtuality uses $\boldsymbol{q} = (\ell, \ell - p_1, \ell - p_1 - p_2, \ell - p_3)$, $p_1^2 = p_2^2 = p_3^2 = 0$, and $v = (p_1 + p_2 - p_3)^2$. With the nonplanar convention for s, t, expanding v gives $2p_2 \cdot p_3 = s + t - v$.

The rational nonplanar profiles are $(s, t) = (-149, -147), (-167, -157)$. The pentagon profiles are $(-51, -53, -57, -59, -63)$ and $(-71, -73, -77, -79, -83)$ in the displayed invariant order. Each rational profile supplies sixteen targets. The symbolic nonplanar, pentagon and virtuality-box tests supply ten, six and eight targets. They start with the corner and single additional powers, followed by auxiliary numerators and single removed denominators where available. These definitions give the 88 targets of the transfer test.

G Master bases and dimensionally consistent target construction

For a complete independent family basis, the number of scalar products involving loop momenta is

$$N_{\mathrm{sp}} = \frac{L(L+1)}{2} + LE. \tag{43}$$

There are L loop squares and $L(L-1)/2$ distinct loop cross-products, followed by LE products with the E independent external momenta. This count supplies a direct rank check for the independent denominator lists. The restricted vacuum parent in Eq. (19) instead uses the explicitly dependent massive denominators and their partial fractions.

For index vector $\boldsymbol{a}$, define the positive-index sum $r = \sum_i \max(a_i, 0)$, numerator rank $s_- = \sum_i \max(-a_i, 0)$ and additional-power count $b = \sum_i \max(a_i - 1, 0)$. If the top sector has t_{top} physical denominators, the initial Kira basis calculation uses seed limits $(r, s_-, b) \le (t_{\mathrm{top}} + 2, 2, 2)$ and probe limits $(t_{\mathrm{top}} + 1, 1, 1)$, component by component. Both include all subsectors. The basis check then increases every limit by one and repeats at a second finite-field point. The original sixteen definitions give the same master labels in all three calculations, for 48 completed basis checks. Such agreement supports stability in the tested domains; it does not prove completeness for all integer indices [4, 5].

The initial sixteen classes give 49 family-local master entries. The tadpole gives one. The four bubble classes give one, two, two and three, in their stated order. The five triangle classes give one, two, four, three and five. The three box classes give three, four and eleven. The massless two-loop two-point, sunrise and vacuum sunset classes give two, three and two. For example, the equal-mass sunrise basis is

$$M_1 = I_{(1,1,0,0,0)}, \qquad M_2 = I_{(1,1,1,0,0)}, \qquad M_3 = I_{(1,1,1,-2,0)}. \tag{44}$$

The first integral separates into two tadpoles, and the last contains the numerator $[(p - \ell_1)^2]^2$. These are basis integrals, whose numerical values need not be calculated to train a reduction policy. Each later family instance retains its own explicit index basis and parameter assignment in the integral data.

The mass dimension of Eq. (1) is $Ld - 2\sum_i a_i$: each loop measure contributes d powers of mass and each inverse propagator contributes two. In the restricted vacuum data, $[V_{\boldsymbol{b}}^{(L)}] = Ld - 2L - 2B$. Thus targets of different total powers can be combined consistently as

$$T = \sum_{\nu} \kappa_\nu d^{\eta_\nu} \mu^{B_\nu - B_*} V_{\boldsymbol{b}_\nu}^{(L)}, \qquad B_* = \max_{\nu} B_\nu. \tag{45}$$

Every term has dimension $Ld - 2L - 2B_*$. The construction uses $\eta_\nu \in \{0, 1\}$ and $\kappa_\nu = n_\nu / q_\nu$, with $n_\nu \in \{-3, -2, -1, 1, 2, 3\}$ and $q_\nu \in \{1, 2, 3\}$. Each expression contains one, two or three distinct roots. Individual roots have $2 \le B \le 5$; Euler seeds have $1 \le B \le 5$, and mass identities extend

through $B = 6$. Each added family contains 128 distinct targets, including individual roots. The identity pools contain 35, 160 or 545 rows for two, three or four massive denominators.

For integer profile number n, the generated mass ratios are $R = 2 + ((n-1) \bmod 3)$, $\rho_1 = 1$ and $\rho_{j+2} = 2 + 3n + j(n+2)$ for $0 \le j \le R-2$. Profiles $n = 1, \ldots, 100$ enter at loop orders 2–12; profiles 101–1350 enter at orders 1–12. Distinct masses make the partial-fraction denominators nonzero. A profile remains in the same training or testing group at every loop order. These rules state the extent of the high-loop collection and its restricted algebraic structure.

G.1 Analytic check of the restricted masters

The recurrences in Appendix D reduce the specified family to $M_j^{(L)}$. We can also derive their values to check this choice. Write $\delta = d/2$, $\alpha = L(\delta - 1)$ and $c = L - (L-1)\delta$. The symbol Γ denotes the Euler gamma function, and a, b below are denominator powers. The standard Euclidean two-denominator convolution follows from Feynman parameters [1]:

$$\begin{aligned} J(a,b;Q) &= \int \frac{d^dk}{\pi^\delta} \frac{1}{(k^2)^a((Q-k)^2)^b} \\ &= (Q^2)^{\delta-a-b} \frac{\Gamma(a+b-\delta)\Gamma(\delta-a)\Gamma(\delta-b)}{\Gamma(a)\Gamma(b)\Gamma(2\delta-a-b)}. \end{aligned} \tag{46}$$

To obtain this expression, combine the denominators with

$$\frac{1}{A^a B^b} = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \int_0^1 \frac{x^{a-1}(1-x)^{b-1}\,dx}{[xA + (1-x)B]^{a+b}}. \tag{47}$$

Shifting k completes the square and gives the radial denominator $k^2 + x(1-x)Q^2$. The radial integral equals $\Gamma(a+b-\delta)[x(1-x)Q^2]^{\delta-a-b}/\Gamma(a+b)$. The remaining beta integral is $\Gamma(\delta - b)\Gamma(\delta-a)/\Gamma(2\delta-a-b)$, which yields Eq. (46).

At fixed Q, successive convolution of the L unit-power massless factors gives

$$C_L (Q^2)^{-c}, \qquad C_L = \frac{\Gamma(c)\Gamma(\delta-1)^L}{\Gamma(\alpha)}. \tag{48}$$

For $L = 1$ the coefficient is one. Applying Eq. (46) with $a = c$, $b = 1$ advances c to $c + 1 - \delta$ and proves the formula by induction. The final integral over Q then gives

$$\begin{aligned} M_j^{(L)} &= \frac{C_L}{\Gamma(\delta)} \int_0^\infty \frac{x^{\alpha-1}\,dx}{x + \mu_j} \\ &= \mu_j^{\alpha-1} \frac{\Gamma(c)\Gamma(\delta-1)^L\Gamma(1-\alpha)}{\Gamma(\delta)}. \end{aligned} \tag{49}$$

The change $x = \mu_j y$ reduces the integral to $\Gamma(\alpha)\Gamma(1-\alpha)$ for $0 < \operatorname{Re}\alpha < 1$. Analytic continuation gives the dimensionally regulated result [1]. Distinct positive fixed mass ratios give distinct exponentials $\rho_j^{L(d/2-1)-1}$ as functions of d. After common factors cancel, a rational-function relation would have a unique largest exponential as real d increases. Rational coefficients grow at most as powers and cannot cancel that term. Removing its zero coefficient and repeating establishes independence over $\mathbb{Q}(d)$ at these fixed mass ratios. Special dimensions and coincident masses require their own analysis.

H Finite minimum-operation checks

A known reverse construction supplies a valid training path, but a minimum also requires a specified operation cost and a complete set of alternatives. We checked two small physical systems

to make this distinction explicit. Define the massless bubble $B_{a,b} = I_{(a,b)}$ with denominators ℓ^2+i0 and $(\ell-p)^2+i0$, and $p^2 = s \neq 0$. Contracting the derivative with ℓ uses $2\ell\cdot(\ell-p) = \ell^2+(\ell-p)^2-s$. Substituting this equality in Eq. (7) gives

$$0 = (d-2a-b)B_{a,b} - bB_{a-1,b+1} + bsB_{a,b+1}. \tag{50}$$

Changing variables $\ell \mapsto p-\ell$ exchanges a, b and gives the second operator identity. Integrals with only a massless denominator vanish in dimensional regularization [1]. The two operators at $(1,1)$, the first at $(2,1)$ and the second at $(1,2)$ therefore give

$$\begin{pmatrix} 0 & s & 0 & d-3 \\ s & 0 & 0 & d-3 \\ d-5 & -1 & s & 0 \\ -1 & d-5 & s & 0 \end{pmatrix} \begin{pmatrix} B_{2,1} \\ B_{1,2} \\ B_{2,2} \\ B_{1,1} \end{pmatrix} = 0. \tag{51}$$

The first two rows yield $B_{2,1} = B_{1,2} = (3-d)B_{1,1}/s$. Substituting in the third yields $B_{2,2} = (d-6)(d-3)B_{1,1}/s^2$.

For the massive tadpole $T_a = \int d^d\ell\,(\ell^2-m^2+i0)^{-a}/(i\pi^{d/2})$, the same derivative gives

$$0 = dT_a - 2a\int \frac{d^d\ell}{i\pi^{d/2}} \frac{\ell^2}{(\ell^2-m^2+i0)^{a+1}} = (d-2a)T_a - 2am^2T_{a+1}. \tag{52}$$

Seeds $a = 1, 2, 3$ reduce T_2, T_3, T_4 to T_1. We keep the master column protected in both systems. An allowed step selects a nonzero pivot in an unprocessed nonmaster column, normalizes its row and eliminates that column from the other rows. The cost counts one division per nonzero nonpivot source entry when normalization is needed, plus one multiplication and one subtraction for each nonpivot source-entry update. It excludes assignments and the internal polynomial operations of a rational-function operation.

Let S be the exact matrix state, $\mathcal{A}(S)$ its allowed actions, $c(S,a)$ the stated step cost and S_a the resulting state. Splitting every complete path into its first action and its remaining path gives the finite recurrence [14]

$$V(S) = \min_{a\in\mathcal{A}(S)} \{c(S,a) + V(S_a)\}, \qquad V(S_{\text{done}}) = 0. \tag{53}$$

Each action removes an unprocessed column. Induction on the number of remaining columns proves the minimum after all actions have been evaluated. Repeating this calculation for every source-row subset of nonmaster rank three also minimizes over initial equation selection.

Our bubble check covers 113 states, 234 transitions and 152 complete paths across five sufficient subsets. Its minimum is seven coefficient operations with subset selection and thirteen with the full pool. The tadpole check covers 24 states, 32 transitions and fifteen paths; its minimum is seven operations with its single sufficient subset. Each pool requires at least three equations. The smallest number of distinct seed labels is two for the bubble and three for the tadpole. Exact replay and six native Kira master-coefficient comparisons agree. These are minima for the displayed identities, protected basis, allowed operations and cost. They do not establish minimum elapsed time or an optimum over all IBP identities.

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